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# Integrating Rooftop Rainwater Harvesting with Zone-Specific Managed Aquifer Recharge in Dhaka: A Recharge Opportunity Index Approach

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Research Title: Integrating Rooftop Rainwater Harvesting with Zone-Specific Managed Aquifer Recharge in Dhaka: A Recharge Opportunity Index Approach

## Introduction

Since the 1990s, rapid urban expansion, extensive paving, loss of open space, and increased abstraction through deep tube wells have steadily eroded Dhaka's natural groundwater recharge, and by the 2010s extraction from its aquifers far outpaced seasonal replacement. Groundwater extraction has exceeded natural replenishment for decades, causing water levels to decline across much of the city [1], [2], [3]. This trend increases energy costs for pumping, threatens long-term water security, and creates greater hardship for households and small businesses that cannot afford deeper extraction systems [1], [2]. Dhaka receives substantial monsoon rainfall, but much of that water is rapidly diverted into drains and canals rather than being captured and allowed to infiltrate. High-density development has also limited the open land needed for conventional recharge ponds or large infiltration systems [1], [2]. Bangladesh has previously considered rooftop rainwater harvesting and artificial recharge as part of its water-security response [4]. However, implementation has remained limited because recharge systems are often proposed without sufficient attention to local soil and groundwater conditions, similar recommendations are applied to neighbourhoods with very different recharge potential, and building regulations lack connected approval, inspection, incentive, and maintenance mechanisms [5].

This research proposes a practical, zone-specific framework for reducing pressure on Dhaka's depleted groundwater system by evaluating where rooftop rainwater harvesting can be connected to Managed Aquifer Recharge (MAR, the deliberate recharge of aquifers via engineered infiltration), which recharge approach is suitable in each priority zone, and what policy mechanisms could make implementation realistic. The study introduces a Recharge Opportunity Index (ROI, defined in the Methodology) to identify locations where rainwater harvesting and MAR are environmentally needed, technically feasible, spatially possible, and administratively implementable. This framework could also generalise to other groundwater-stressed cities.

## Research Questions and Objectives

**Main research question:** How can rooftop rainwater harvesting be integrated with zone-specific Managed Aquifer Recharge to identify practical, high-priority groundwater-resilience interventions in Dhaka?

**Subquestions:**

1. Which selected Dhaka zones experience the greatest combined groundwater stress and urban water-security pressure?
2. How much rooftop runoff could realistically be captured and infiltrated in those priority zones?
3. Which MAR approaches and policy arrangements are most feasible under the hydrogeological, land-use, water-quality, and infrastructure constraints of each zone?

**Objectives:**

a. To identify priority Dhaka zones based on groundwater stress and rooftop-MAR opportunity.
b. To estimate realistic rooftop runoff capture and recharge potential.
c. To evaluate suitable MAR techniques for each priority zone.
d. To develop a policy framework supporting implementation and maintenance.

## Literature Review and Research Gap

Previous studies have established that Dhaka's groundwater decline is primarily linked to prolonged abstraction and reduced natural recharge in an increasingly urbanised landscape [1], [2], [3]. Specifically, [1] and [2] document the extent and drivers of level decline, while [3] extends this through MODFLOW-based simulation, projecting that continued extraction without major intervention could deepen water-level decline and raise the cost of urban water supply. Hydrogeological investigation of the upper Dupi Tila aquifer [6] complements this citywide evidence, identifying potential MAR approaches for Dhaka, including recharge trenches, recharge pits, soil-aquifer treatment, aquifer storage and recovery, aquifer storage transfer and recovery, and wetland-based recharge, and showing that suitability varies by local geological conditions and urban land use. While the citywide modelling in [3] projects continued decline, the local hydrogeological analysis in [6] suggests engineered recharge could offset losses in favourable zones, pointing to a need for zone-specific assessment. Research on rooftop rainwater harvesting in Bangladesh [7] has demonstrated its potential as an alternative source of safe drinking water, particularly in arsenic-affected and water-scarce areas, and the Bangladesh National Building Code 2020 has separately recognised rainwater harvesting as a possible response to groundwater depletion [5]. Recent GRACE/GLDAS downscaling methods [8] offer complementary regional evidence not yet integrated with rooftop-harvesting or MAR feasibility work. Yet adoption remains limited, especially where systems require upfront investment, technical design support, inspection, maintenance, and secure connection to approved recharge facilities.

Research gap: Existing studies generally examine groundwater depletion, MAR feasibility, rooftop rainwater harvesting, or policy compliance separately. Few studies combine these dimensions, groundwater stress, runoff potential, hydrogeological suitability, urban constraints, water-quality risk, and institutional feasibility, at a neighbourhood scale for Dhaka. Much of this evidence also predates recent satellite and modelling tools, and few studies pair engineering feasibilities with policy design in one method. This study addresses that gap through a Recharge Opportunity Index and an associated policy roadmap.

## Proposed Methodology

The method follows a mixed-methods design consisting of spatial analysis, water-balance estimation, MAR suitability assessment, document review, and an optional demonstration unit, combining objective spatial and hydrological measurement with qualitative policy input, since recharge opportunity requires both data and judgement.

### 1. Study Area and Data Collection

The study will focus on selected Dhaka zones representing different groundwater-stress and urban-development conditions, such as Tejgaon, Mirpur, Malibagh, and one additional comparison zone selected after initial data review. For this study, a 'zone' will mean a defined GIS analysis unit based on administrative boundaries and data availability, with the final units selected before index calculation. Zones are selected purposively to represent contrasting stress and development conditions, so the four-zone sample favours contextual insight over citywide generalisability. Data will be compiled from the following sources:

a. Groundwater-level or piezometer records (Bangladesh Water Development Board, Dhaka WASA, published research) [1], [2], [3]
b. Rainfall data from the Bangladesh Meteorological Department
c. Building footprint, roof-area, land-use, drainage, and open-space data from RAJUK planning materials, satellite imagery, and GIS layers
d. Hydrogeological information on aquifer materials and infiltration constraints [6]
e. Relevant Building Code and RAJUK approval documents
f. Informal, anonymised input from building owners, engineers, planners, and water-service personnel, given with consent

## 2. Rooftop Runoff and Recharge Estimation

Potential annual rooftop runoff will be estimated as $V_{runoff} = P \times A \times C$, where P is annual rainfall depth, A is rooftop catchment area, and C is the runoff coefficient. A more realistic usable volume will then be estimated after accounting for first-flush diversion, leakage, overflow, filtration losses, storage limitations, and infiltration efficiency: $V_{usable} = V_{runoff} \times (1 - L_f) \times \eta_s$. The study will distinguish between theoretical rooftop runoff, captured runoff, water delivered to the recharge unit, estimated infiltration volume, and plausible contribution to groundwater recharge, with sensitivity analysis testing how results change as rainfall, roof area, runoff coefficient, loss fraction, and infiltration efficiency vary.

## 3. Recharge Opportunity Index Development

Each component score is built from a defined indicator and normalised to 0 to 100 across the selected zones using min-max scaling, so the highest-scoring zone on that indicator receives 100 and the lowest receives 0. Because scores are relative to this four-zone sample rather than fixed absolute thresholds, a score of 0 marks is the least stressed or least suitable zone among those studied, not a safe or low-priority condition in absolute terms. Where an indicator is missing for a zone, that zone receives the sample median, disclosed in the results. For components with more than one indicator, the normalised indicator scores are averaged with equal weight to produce that component score. Table 1 defines each component.

| Component | Indicator(s) | Scoring direction |
|---|---|---|
| G, groundwater stress | Depth to water table; annual rate of decline | Higher stress/decline scores higher (greater priority) |
| R, rooftop runoff potential | Estimated usable rooftop runoff, Vusable | Higher usable volume scores higher |
| H, hydrogeological suitability | Soil permeability; aquitard thickness; aquifer depth | Higher suitability for infiltration scores higher |
| U, urban/infrastructure feasibility | Available open space; distance from foundations and utilities | Higher feasibility scores higher |
| Q, water-quality safety | Contamination-screening result; pretreatment feasibility | Higher safety (lower contamination risk) scores higher, not the reverse |
| I, policy/institutional readiness | Building Code compliance status; maintenance capacity | Higher readiness scores higher |

Each selected zone will receive a standardised score between 0 and 100, calculated as ROI = 0.30G + 0.20R + 0.20H + 0.15U + 0.10Q + 0.05I, where G = groundwater-stress score, R = rooftop runoff potential score, H = hydrogeological suitability score, U = urban-space and infrastructure feasibility score, Q = water-quality safety score, and I = policy and implementation readiness score. The initial weights are theory-informed and will be tested through equal-weight and alternative-weight sensitivity scenarios. The index will therefore be interpreted as a transparent screening tool rather than a definitive measurement of recharge potential. A zone will not receive high priority on groundwater stress alone but must also show realistic rooftop-collection and recharge opportunity.

## 4. MAR Technique Selection

Potential recharge techniques will be screened using a decision matrix based on shallow soil and sediment permeability, target-aquifer depth and characteristics, available open space and distance from building foundations, proximity to drains, utilities, roads, and contaminated sites, runoff quantity and quality, pretreatment needs, and capital, maintenance, and monitoring requirements. Rooftop runoff connected to filtered recharge pits or trenches is expected to be the most broadly applicable option [4], [6]. Soil-aquifer treatment will be considered where additional water-quality protection is needed, while aquifer storage and recovery (ASR) or aquifer storage transfer and recovery (ASTR) will only be recommended where data justify injection-based recharge [6]. The study will explicitly distinguish between recharge suitability and drinking-water suitability, and will not recommend recharge near-known contamination sources without additional safeguards.

## 5. Policy Analysis and Optional Demonstration

The study will review the policy journey from design approval, installation, and inspection through to maintenance responsibility, water-quality monitoring, and enforcement, and will develop a policy roadmap recommending standard designs, compliance checks, incentives, technical guidance, maintenance certification, and a GIS-based registry of approved systems for public buildings, new developments, and existing buildings. If a safe and suitable building site is available, a small demonstration system (rooftop → gutter → first-flush diverter → sand/gravel filter → recharge pit) may be constructed or simulated to measure rainfall depth, roof area, estimated and captured runoff, turbidity, pH, and infiltration rate. This demonstration is optional and secondary to the main screening and policy analysis.

# Expected Outcomes

Results are expected to provide:

1. A map of priority Dhaka zones by groundwater stress and rooftop-MAR opportunity
2. A Recharge Opportunity Index and technique-selection matrix for priority zones
3. Estimated annual rooftop runoff and plausible infiltration potential under transparent assumptions
4. A practical MAR recommendation for each priority zone
5. A policy roadmap identifying institutions, enforcement stages, incentives, and maintenance mechanisms

6. A replicable framework adaptable to other dense, groundwater-stressed cities

## Potential Limitations

The project will rely partly on published groundwater data because continuous, high-resolution local piezometer records may not be available. The study will not develop a new GRACE/GLDAS downscaling model but will use published or accessible datasets as supplementary regional evidence [8]. A short-duration demonstration cannot prove long-term aquifer recharge or citywide groundwater recovery, and policy findings will rely on document review and limited stakeholder input rather than a large formal institutional survey.

## Project Practicalities

The project will take 18 weeks, organised as follows:

| Weeks | Activity |
|---|---|
| **Weeks 1–3** | Literature and policy review |
| **Weeks 4–6** | Data collection and preparation |
| **Weeks 7–9** | GIS mapping and index design |
| **Weeks 10–11** | MAR suitability assessment |
| **Weeks 12–14** | Runoff modelling, sensitivity analysis, and optional pilot monitoring |
| **Weeks 15–16** | Policy and cost analysis |
| **Weeks 17–18** | Final reporting, maps, and presentation |

The indicative budget totals approximately BDT 34,500 to 47,500, covering data collection and site visits, GIS and data preparation, optional demonstration materials, basic water-quality testing, documentation and presentation, and contingency. The optional demonstration, if undertaken, will be limited to a micro-bench-scale model or simulation using secondary data so that total costs remain within the indicated budget range.

Beyond the data-access limitation noted above, key risks include restricted stakeholder availability, addressed with broader outreach, and schedule slippage, covered by the final-week buffer. Interviews, where conducted, will be voluntary and anonymised.

## Post-program Plan

Following the study, the Recharge Opportunity Index will be refined with stakeholder feedback from Dhaka WASA and RAJUK, and the framework will be prepared for publication and potential pilot implementation in a priority zone.

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